
TEKLE MAHRAY BAHTA

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PROFESSIONAL SUMMARY

Dedicated and analytical Computer Science student at Full Sail University with strong expertise in C++ systems programming, memory management, and binary file serialization. Experienced in designing modular game engines, implementing defensive error handling, and conducting computational linguistics research. Adept at utilizing Git version control, GitHub project management, and collaborative development workflows.

TECHNICAL SKILLS

Languages: C++, C, Assembly (6502)

Tools & Frameworks: Visual Studio 2022 & 2026, CMake, Git, GitHub Pages, Markdown, Obsidian

Core Competencies: Object-Oriented Programming, Memory Management (Rule of Three), Binary/Text Serialization, Pointer Manipulation, Bitfields, Algorithmic Problem Solving, Computational Philology

PROJECTS

Sovereign Frequencies (Tigrinya Linguistic Game Engine) | C++, Visual Studio, Git

- Developed a multi-phase C++ console engine integrating structural core classes, option generation, and game flow state management.
- Implemented text (`roots.txt`) and binary (`roots.dat`) serialization protocols with unique ID tracking and strict all-or-nothing file validation to prevent partial or corrupted saves.
- Designed hardened input validation utilities to prevent stream failures and runtime exceptions during user menu interactions.

- Applied terminal styling enhancements to structure headings, prompts, and success/error states clearly and consistently.
- Managed iterative development using GitHub repositories, branching workflows, issues, and milestone tracking boards.

C++ Systems Programming Labs & Renderers | C++, CMake, Win32

- Implemented pointer manipulation, custom bitfield structures, and Rule-of-Three memory management across systems programming labs.
 - Built rendering applications featuring spatial raycasting, dynamic allocation, and interactive geometry picking.
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EDUCATION

Full Sail University — Winter Park, FL

Bachelor of Science in Computer Science

Relevant Coursework: Systems Programming, Project & Portfolio, Software Development, Data Structures and Algorithms
